Product Requirement Document

Feature Title: Device Filtering, Cart Management & Checkout Simulation **Document Version:** 2.1 **Platform:** bstackdemo.com (Web & Mobile)

Feature Overview

The purpose of this feature set is to simulate a responsive e-commerce experience for the BrowserStack Demo Application. This enables QA engineers and developers to test browser compatibility across critical user journeys. This feature will allow users to:

1. **Filter** the product grid by specific device vendors (e.g., Apple, Samsung).
2. **Sort** displayed products by price (Ascending/Descending).
3. **Add items** to the cart directly from the product listing.
4. **Manage Cart** contents via a dynamic flyout modal (adjust quantities, remove items).
5. **Authenticate** via a simulated login gate during checkout.
6. **Complete Order** through a shipping address form and receive confirmation.

Key Functional Requirements

1. Filter Products by Vendor

- **UI Component:** A persistent sidebar on Desktop (left-aligned) or a collapsible filter menu on Mobile.
- **Functionality:** Users can select one or multiple vendors to filter the grid.
- **Vendor Options:**
  - Apple
  - Samsung
  - Google
  - OnePlus
- **Behavior:** The product grid refreshes instantly (Client-side filtering) without a page reload.
- **Count:** The number of available devices per vendor should be displayed next to the vendor name (e.g., "Apple (9)").

2. Product Listing & Sorting

- **Grid Layout:** Products are displayed in a responsive grid (4 columns on Desktop, 2 on Tablet, 1 on Mobile).

- **Product Card Details:**
  - **Image:** High-resolution device thumbnail.
  - **Title:** Full device model name (e.g., "iPhone 12 Pro Max").
  - **Price:** Formatted in USD (e.g., "$1099").
  - **Action:** "Add to cart" button.
- **Sorting Mechanism:**
  - A dropdown menu positioned at the top-right of the grid.
  - **Options:** "Lowest to Highest" and "Highest to Lowest."
  - **Default State:** Default sorting is by relevance/popularity.

3. Add Item to Cart

- **Interaction:** Clicking "Add to cart" triggers a non-blocking action.
- **Feedback:** The button state creates a visual feedback loop (e.g., brief loading spinner or text change).
- **Cart Counter:** The floating cart icon (top-right header) updates its numeric badge immediately (e.g., increments from 0 to 1).

4. Manage Cart ("Review Bag")

- **Access:** Clicking the cart icon opens a **Cart Flyout Modal** from the right side of the screen (overlaying the current page).
- **Content Display:**
  - List of all added items with thumbnails.
  - **Quantity Control:** Users can increase (`+`) or decrease (`-`) the count of specific items.
  - **Deletion:** Decreasing quantity to 0 removes the item from the cart.
- **Pricing Summary:**
  - **Subtotal:** Dynamically calculated based on `(Price x Quantity)` for all items.
  - **Checkout Button:** A persistent footer button within the modal to proceed.

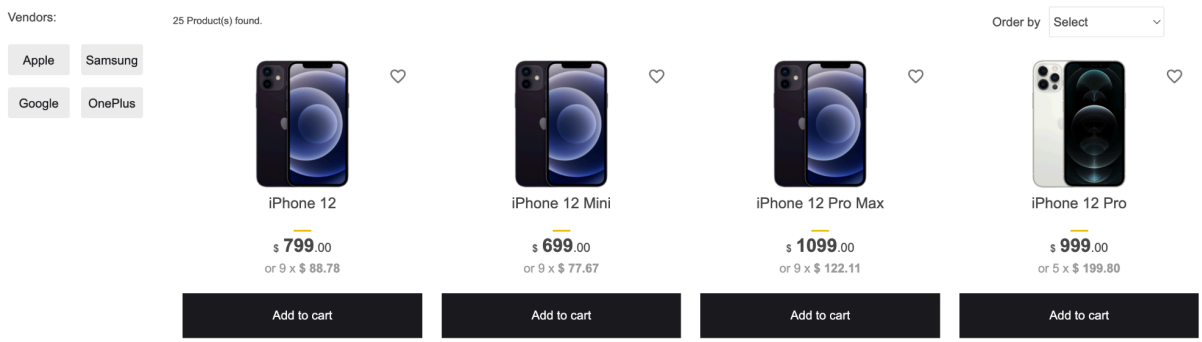
5. Checkout & Authentication Flow

- **Login Gate:** Clicking "Checkout" checks for an active user session.
  - *If inactive:* Redirects to `/signin`.
  - *Credentials:* Accepts `demouser` / `testingisfun` (and `image_not_loading_user` for negative testing).
- **Shipping Details:**
  - A form requiring: First Name, Last Name, Address, State, and Postal Code.
  - **Validation:** Basic field validation (non-empty fields).
- **Order Confirmation:**
  - Displays a success message: "Order has been successfully placed."
  - Provides a generated Order ID for reference.
  - Button to "Continue Shopping" which resets the flow.

Sample Workflows

1. Search/Filter and Add Single Item

1. **Navigate** to the bstackdemo homepage.
2. **Locate** the "Vendor" sidebar on the left.



3. **Select** "Google" to filter results.
4. **Click** "Add to cart" on the "Google Pixel 5".
5. **Verify** the Cart Icon badge updates to "1".

2. Multi-Vendor Selection and Cart Modification

1. **Filter** by "Apple" and add "iPhone 12 Mini" to the cart.
2. **Uncheck** "Apple" and check "OnePlus".
3. **Add** "OnePlus 9" to the cart.
4. **Open** the Cart Flyout (Click Bag Icon).
5. **Adjust** quantity of "iPhone 12 Mini" to 2 using the + button.
6. **Verify** the Subtotal updates automatically to reflect the price of 3 total devices.

3. Complex Workflow: Unauthenticated Checkout

Objective: Add high-value items, attempt checkout, login, and complete purchase.

1. **Sort** products by "Highest to Lowest."
2. **Add** "iPhone 12 Pro Max" ($1099) and "OnePlus 9" ($729) to cart.
3. **Click** the Cart Icon.
4. **Click** "Checkout."
    - *System redirects to Login Page.*
5. **Enter Credentials:**
    - Username: `demouser`
    - Password: `testingisfun`

6. **Click** "Log In."
    - *System redirects to Shipping Page (Cart is preserved).*
7. **Enter Shipping Info:**
    - First Name: "Jane"
    - Last Name: "Doe"
    - Address: "1 Infinite Loop"
    - State/Zip: "CA", "95014"
8. **Click** "Submit."
9. **Verify** "Order Placed" screen and Order ID generation.

Acceptance Criteria

- **Filtering Accuracy:** Selecting a vendor must strictly display *only* devices from that manufacturer.
- **Pricing Logic:** The subtotal in the cart must exactly match the sum of individual item prices multiplied by their respective quantities.
- **Session Persistence:** If a user adds items to the cart and then refreshes the browser, the items should remain in the cart (Local Storage check).
- **Authentication Gate:** No user should be able to reach the "Order Placed" screen without passing through the Login screen first.
- **Responsive Design:** The Cart Flyout must be usable on mobile devices (covering the full screen if necessary) without breaking layout.

Dependencies

- **Product Data:** Hardcoded JSON data containing device names, prices, and image URLs.
- **Auth Service:** A mock authentication service that validates the specific string pair `demouser`/`testingisfun`.
- **Browser Storage:** Functionality depends on `window.localStorage` availability to persist cart state across the login redirect.